

AD 502 – Machine Learning

Pre-Reading Material – Lecture 6

Scope, Limitations and Challenges of Machine Learning

Estimated Reading Time: 25–30 minutes

Reference Reading

1. Malik, M. M. (2020). *A Hierarchy of Limitations in Machine Learning*. Berkman Klein Center for Internet & Society at Harvard University.
2. Mukhamediev, R. I., et al. (2022). *Review of Artificial Intelligence and Machine Learning Technologies: Classification, Restrictions, Opportunities and Challenges*. Mathematics.

Why This Topic Matters

While machine learning (ML) has proven incredibly useful for prediction in social and technical systems, the field is increasingly grappling with how these models are "wrong". Understanding the **mathematical, philosophical, and sociological limitations** of ML is essential for both developers and consumers to identify potential failure points and prevent unintended harms, such as biased healthcare decisions or fragile financial models.

Learning Outcomes

After completing this pre-reading, you should be able to:

- Identify the diverse **sub-domains** and applications within the scope of AI and ML.



- Explain the **Hierarchy of Limitations**, ranging from quantification to model evaluation.
- Distinguish between **internal (technological)** and **external (social/organizational)** restrictions on ML adoption.
- Describe how **non-causal correlations** and the **central tendency** can lead to model failures.

Activate Your Prior Knowledge

1. The "Cat-ness" Problem

Suppose you train a model to recognize cats. It achieves 99% accuracy on pixels.

Think: Does the model actually understand the "construct" of a cat, or is it just identifying pixel patterns that happen to correlate with human labels?

2. The Average Pilot

The Air Force once designed seats for the "average pilot," only to find the seat fit *zero* actual pilots.

Think: When an ML model optimizes for the "central tendency" (the mean or majority), who gets left behind?

3. Trusting the "Black Box"

A model predicts a patient needs a specific treatment but cannot explain why.

Think: Is high accuracy enough to justify using a model if we don't understand its internal decision-making?

Prerequisite Concepts

You should be familiar with:

- **Supervised vs. Unsupervised Learning**



- **Correlation vs. Causality**
- **Cross-validation and Generalizability**
- **Feature Extraction**

Core Concepts

1. The Scope of Machine Learning

ML is the core of modern Artificial Intelligence, encompassing several scientific areas:

- **Computer Vision:** Image recognition and processing.
- **Natural Language Processing (NLP):** Machine translation, sentiment analysis, and text generation.
- **Robotics:** Learning in soft robotics and autonomous systems.
- **Specialized Domains:** Applications in healthcare (genomics), agriculture (soil salinity), and resource extraction (mining).

2. The Hierarchy of Limitations

Momin Malik proposes a four-level hierarchy of decisions that limit ML reliability:

1. **Quantification:** Choosing quantitative over qualitative analysis sacrifices narrative meaning and lived experience. We often work with **proxies** (like test scores) instead of actual **constructs** (like intelligence).
2. **Probability-based Modeling:** ML relies on the **central tendency**, which prioritizes the "average" and punishes outliers or individuals.
3. **Predictive vs. Explanatory:** ML focuses on **prediction** (correlations), which can be fragile, gameable (Goodhart's Law), and unable to guide real-world interventions compared to **causal explanation**.



4. **Cross-validation:** Relying solely on cross-validation can be overly optimistic if there are unmodeled **dependencies** in the data (e.g., temporal or network connections).

3. Internal and External Restrictions

The challenges to ML implementation are divided into two categories:

- **External Restrictions:** Organizational (lack of strategy), Personnel (shortage of data scientists, fear of technology), Economic (high costs), and Social (legal/ethical issues).
- **Internal Limitations:** Data (lack of labeled data, privacy), Learning Processes (computational power requirements), and Results (lack of **interpretability** or the "Black Box" problem).

4. Interpretability vs. Explainability

A major challenge is **Explainable AI (XAI)**. Modern deep learning models are often "Black Boxes". While we can use "post-hoc" methods to explain a single prediction (Local) or the overall model (Global), these interpretations may not reflect the actual causal reality of the system.

Guided Reading Questions

1. What is the "**McNamara fallacy**", and how does it relate to ML's reliance on measurability?
2. Explain why a model that predicts well might still fail when the environment changes (e.g., Google Flu Trends).
3. What are the three major threats to the validity of cross-validation mentioned in the sources?

4. Why are "**resource-based economies**" (like Kazakhstan) facing unique challenges in adopting ML?

Reflection Activity

Write 100–150 words on a scenario where an ML model might provide a "correct" statistical prediction that is morally or practically "wrong."

Consider: The use of credit scores for insurance, risk assessments in criminal justice, or health care cost-optimization.

Self-Assessment

1. In the hierarchy of limitations, problems with **quantification** propagate downstream to affect all other levels. (True/False)
2. **Goodhart's Law** states that when a measure becomes a target, it ceases to be a good measure. (True/False)
3. The extra loss incurred by an online learner compared to a fixed parameter choice is called _____.
4. ML models that can provide their own faithful explanations are called _____ models.

Key Takeaway

The scope of ML is vast, but its success is bound by its **mathematical and social assumptions**. To be responsible practitioners, we must look beyond "accuracy" and consider the **construct validity, causal reliability, and social impact** of our models.

A model is only as meaningful as the way the world was quantified to create it.

